

What is RAG? (Retrieval-Augmented Generation)

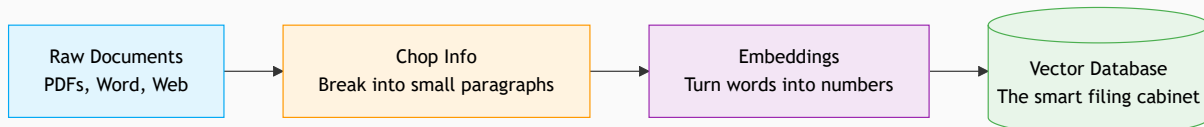
The Analogy: Open-Book vs. Closed-Book Test

Imagine giving an AI an exam. Normally, it takes a "closed-book" test, replying entirely from its memory. Sometimes it forgets facts or confidently guesses the wrong answer (which we call a hallucination).

RAG turns this into an *open-book test*. It gives the AI a search engine so it can look through your specific documents to find the exact, correct facts before it writes an answer.

Phase 1: Preparation (Building the Library)

Before the AI can answer anything, we have to prepare your files so a computer can easily search them. We don't just drop whole 500-page books into the system; we have to organize them.



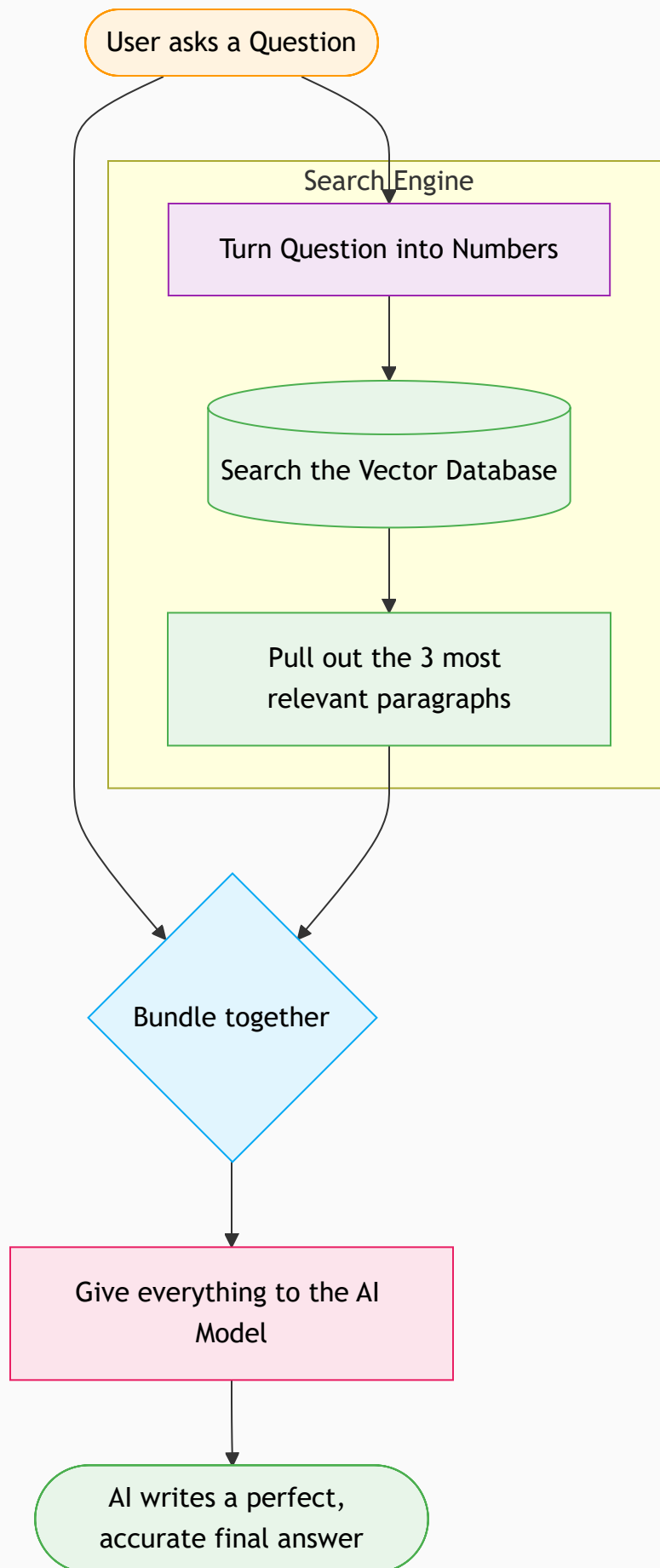
1. Chop into chunks: We take your big documents and slice them into small, bite-sized pieces, like individual paragraphs.

2. Turn Words into Numbers (Embeddings): Computers understand numbers better than text. We use a tool that translates the *meaning* of the text into a long list of numbers. If two paragraphs mean the same thing, their numbers will look similar.

3. Save in a Cabinet (Vector Database): We store all those numbers in a special database that is designed specifically to quickly compare and match similar concepts.

Phase 2: Question Time (Retrieval & Answering)

Now the library is built and ready. Watch what happens when a user actually asks a question.



1. Translate the Question: Just like we did with the documents, we translate the user's plain-English question into numbers.

2. Search (Retrieval): The database quickly compares the question's numbers against all the paragraphs we stored earlier. It pulls out the 3 or 4 paragraphs that have the closest matching meaning.

3. Bundle & Generate: We bundle the user's original question AND the newly found paragraphs together into a single package. We hand this to the AI. Because the AI can now read the exact facts right there on the spot, it simply summarizes them into a perfect final answer.